Data and Visual Analytics

ASSIGNMENT - 2

Mentor :

P.Srikanth

Done By :

Karnika - 160123729007

J.Parna Sree - 160123729008

Aim: To design and implement a complete data analysis pipeline involving data extraction, preprocessing, feature engineering, text vectorization, dimensionality reduction, evaluation metrics, and visualization, in order to analyze user interactions and content patterns in an online social network dataset.

Workflow :

- **Data Generation** – Created large-scale multilingual dataset
- **Preprocessing** – Cleaned captions (removed noise like hashtags, mentions)
- **Exploratory Data Analysis (EDA)** - Language distribution analysis , Caption length statistics
- **Vectorization** – TF-IDF, Bag of Words, and BERT embeddings
- **Dimensionality Reduction** – PCA and t-SNE for visualization
- **Classification** – Logistic Regression, Naive Bayes, Random Forest, ANN
- **Prediction** – Language detection on new inputs
- **Graph Visualization** – Social network analysis using NetworkX

Dataset Description:

A synthetic dataset containing 500,000 records is generated in JSON format, where each record includes:

- User Profile Data (user ID, location, followers, following count)
- Content Data (post ID, multilingual captions in English and Indic languages)
- Interaction Data (hashtags, mentions/tags)
- Network Relationships (friend connections, suggestions)
- Engagement Metrics (likes, comments, shares)

Step 1: Data Generation

- Generated 500,000 synthetic records
- Included English + Indic languages

```
!nvidia-smi
```

```
import torch
if torch.cuda.is_available():
    device = torch.device('cuda')
    print(f'GPU is connected: {torch.cuda.get_device_name(0)}')
else:
    device = torch.device('cpu')
    print('GPU NOT AVAILABLE. Please enable GPU in Colab.')
```

OUTPUT:

```
Tue Apr  7 10:03:43 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.82.07              Driver Version: 580.82.07      CUDA Version: 13.0     |
```

```
| NVIDIA-SMI 580.82.07          Driver Version: 580.82.07     CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |         Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   39C    P8              10W /   70W |      0MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI          PID   Type   Process name                  GPU Memory |
|        ID   ID                                                   Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
GPU is connected: Tesla T4
Tue Apr  7 10:03:52 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.82.07          Driver Version: 580.82.07     CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |         Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   39C    P8              10W /   70W |      3MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI          PID   Type   Process name                  GPU Memory |
|        ID   ID                                                   Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
GPU is connected: Tesla T4
!pip install -q transformers datasets evaluate jiwer rouge-score networkx matplotlib seaborn
!pip install -q indic-nlp-library
!pip install -q nltk diff-match-patch asr-evaluation sentencepiece scikit-learn
```

OUTPUT:

```
 Preparing metadata (setup.py) ... done
```

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 84.1/84.1 kB 4.1 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.2/3.2 MB 82.8 MB/s eta
0:00:00
  Building wheel for rouge-score (setup.py) ... done
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 40.3/40.3 kB 1.9 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 7.7/7.7 MB 37.4 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 121.1/121.1 kB 7.0 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 43.3/43.3 kB 3.1 MB/s eta
0:00:00
import json
import random
import datetime
import uuid
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import networkx as nx
from sklearn.decomposition import PCA
import warnings
warnings.filterwarnings('ignore')
sns.set_theme(style="whitegrid")
# Define the English + 10 Indic languages
LANGUAGES = ['en', 'te', 'hi', 'ml', 'ta', 'kn', 'mr', 'gu', 'pa', 'or']

LANGUAGE_TEXTS = {

    'en': ["Loving the weather today!", "Just finished a great book.", "Beautiful sunset!", "Delicious meal with friends.", "Working hard today."],

    'hi': ["आज का मौसम बहुत अच्छा है!", "अभी एक बेहतरीन किताब खत्म की।", "सुंदर सूर्यास्त!", "दोस्तों के साथ स्वादिष्ट भोजन।", "आज कड़ी मेहनत कर रहा हूँ।"],

    'te': ["ఈ రోజు వాతావరణం చాలా బాగుంది!", "ఇప్పుడే ఒక గొప్ప పుస్తకం పూర్తి చేసాను.", "అందమైన సూర్యాస్తమయం!", "స్నేహితులతో రుచికరమైన భోజనం.", "ఈ రోజు కష్టపడి పనిచేస్తున్నాను."],

    'ta': ["இன்று வானிலை அருமை!", "ஒரு சிறந்த புத்தகத்தை வாசித்து முடித்தேன்.", "அழகான சூரிய அஸ்தமனம்!", "நண்பர்களுடன் உணவு.", "கடுமையாக உழைக்கிறேன்."],

    'ml': ["ഇന്ന് നല്ല കാലാവസ്ഥയാണ്!", "നല്ലൊരു പുസ്തകം വായിച്ചു.", "മനോഹരമായ സൂര്യാസ്തമയം!", "കൂട്ടുകാരൊത്ത് ഭക്ഷണം.", "കഠിനാധ്വാനം ചെയ്യുന്നു."],

    'kn': ["ಇಂದು ಹವಾಮಾನ ಚೆನ್ನಾಗಿದೆ!", "ಒಂದು ಒಳ್ಳೆಯ ಪುಸ್ತಕ ಓದಿದೆ.", "ಸುಂದರ ಸೂರ್ಯಾಸ್ತ!", "ಸ್ನೇಹಿತರೊಂದಿಗೆ ಊಟ.", "ಕಠಿಣ ಕೆಲಸ ಮಾಡುತ್ತಿದ್ದೇನೆ."],

    'mr': ["आज हवामान खूप छान आहे!", "नुकतेच एक उत्तम पुस्तक वाचून पूर्ण केले.", "सुंदर सूर्यास्त!", "मित्रांसोबत जेवण.", "आज खूप काम करत आहे."],

    'gu': ["આજે હવામાન ખૂબ સરસ છે!", "હમણાં જ એક સરસ પુસ્તક વાંચી પૂરી કરી.", "સુંદર સૂર્યાસ્ત!", "મિત્રો સાથે ભોજન.", "આજે સખત કામ કરી રહ્યો છું."],
```

```
    'pa': ["ਅੱਜ ਮੌਸਮ ਬਹੁਤ ਵਧੀਆ ਹੈ!", "ਹੁਣੇ ਇੱਕ ਵਧੀਆ ਕਿਤਾਬ ਪੂਰੀ ਕੀਤੀ।", "ਸੁੰਦਰ ਸੂਰਜ ਡੁੱਬਣ!", "ਦੋਸਤਾਂ ਨਾਲ ਖਾਣਾ।", "ਅੱਜ ਸਖਤ ਮਿਹਨਤ ਕਰ ਰਿਹਾ ਹਾਂ।"],

    'or': ["ଆଜି ପାଗ ବହୁତ ଭଲ ଅଛି!", "ବର୍ତ୍ତମାନ ଗୋଟିଏ ଭଲ ବହି ପଢିଲି।", "ସୁନ୍ଦର ସୂର୍ଯ୍ୟାସ୍ତ!", "ସାଙ୍ଗମାନଙ୍କ ସହ ଖାଦ୍ୟ।", "ଆଜି ବହୁତ କାମ କରୁଛି।"]

}
NUM_SAMPLES = 500000

data_records = []

print(f"Starting data extraction/simulation for {NUM_SAMPLES} samples...")
for i in range(NUM_SAMPLES):
    lang = random.choice(LANGUAGES)
    user_id = f"user_{random.randint(1000, 99999)}"


    record = {
        "user_profile": {
            "userid": user_id,
            "user_name": f"user_name_{user_id}",

                "location": random.choice(["Mumbai", "Delhi", "Bangalore", "Hyderabad", "Chennai", "Kolkata", "Pune", "Ahmedabad", "Jaipur", "Lucknow"]),

            "followers": random.randint(10, 100000),
            "following_count": random.randint(10, 1000)
        },
        "content_data": {
            "post_id": str(uuid.uuid4()),
            "text_caption": random.choice(LANGUAGE_TEXTS[lang]) + f" #{lang} #socialmedia",
            "language": lang
        },
        "interaction_data": {
            "tags": [f"@user_{random.randint(1000, 99999)}" for _ in range(random.randint(0, 3))],
            "hashtags": [f"#{lang}", "#instadaily", "#trending"]
        },
        "network_relationship_data": {
            "follow_friend_connection": [f"user_{random.randint(1000, 99999)}" for _ in range(random.randint(0, 5))],
            "suggestions": [f"user_{random.randint(1000, 99999)}" for _ in range(random.randint(0, 5))]
        },
        "engagement_metrics": {
            "no_of_likes_count": random.randint(0, 5000),
            "no_of_comments_count": random.randint(0, 500),
            "no_of_share_counts": random.randint(0, 1000)
        }
    }
    data_records.append(record)

file_name = 'instagram_extracted_data.json'
with open(file_name, 'w', encoding='utf-8') as f:
```

```
    json.dump(data_records, f, ensure_ascii=False)

print(f"Successfully extracted and saved {NUM_SAMPLES} samples to {file_name}")
```

OUTPUT:

Starting data extraction/simulation for 500000 samples...

Successfully extracted and saved 500000 samples to instagram_extracted_data.json

Step 2: Data Loading & Exploration

- Loaded JSON dataset
- Verified structure and sample data

Code:

```
with open('instagram_extracted_data.json', 'r', encoding='utf-8') as f:
    loaded_data = json.load(f)

print(f"Total samples loaded: {len(loaded_data)}")
subset_data = loaded_data[:1000]

print("Printing first 2 sample records:\n")
for sample in subset_data[:2]:
    print(json.dumps(sample, indent=2, ensure_ascii=False))
    print("-" * 50)
```

OUTPUT:

Total samples loaded: 500000

Printing first 2 sample records:


```
{
  "user_profile": {
    "userid": "user_70823",
    "user_name": "user_name_user_70823",
    "location": "Lucknow",
    "followers": 49208,
    "following_count": 552
  },
  "content_data": {
    "post_id": "110679cb-ab7a-4527-b22a-8b398ba8183e",
    "text_caption": "Beautiful sunset! #en #socialmedia",
    "language": "en"
  },
  "interaction_data": {
    "tags": [
      "@user_34723",
      "@user_6142",
      "@user_92538"
    ],
    "hashtags": [
      "#en",
```

"#instadaily",
"#trending"
]
},
"network_relationship_data": {
"follow_friend_connection": [
"user_40347",
"user_14538",
"user_93219"
],
"suggestions": [
"user_55135",
"user_62774",
"user_26260",
"user_92509"
]
},
"engagement_metrics": {
"no_of_likes_count": 3036,
"no_of_comments_count": 174,
"no_of_share_counts": 827
}
}

{
"user_profile": {
"userid": "user_76787",
"user_name": "user_name_user_76787",
"location": "Lucknow",
"followers": 64616,
"following_count": 222
},
"content_data": {
"post_id": "082866b0-62ca-4bbc-aa8b-567131a697f0",
"text_caption": "കഠിനാധ്വാനം ചെയ്യുന്നു. #ml #socialmedia",
"language": "ml"
},
"interaction_data": {
"tags": [
"@user_91318"
],
"hashtags": [
"#ml",
"#instadaily",

```
    "#trending"
   ]
  },
  "network_relationship_data": {
   "follow_friend_connection": [
    "user_50835"
   ],
   "suggestions": [
    "user_33040",
    "user_18405"
   ]
  },
  "engagement_metrics": {
   "no_of_likes_count": 557,
   "no_of_comments_count": 133,
   "no_of_share_counts": 609
  }
}
```


Step 3: Exploratory Data Analysis (EDA)

- Language distribution analysis
- Caption length statistics
- Engagement metrics distribution
- Followers vs Engagement insights

```
print("\n===== EXPLORATORY DATA ANALYSIS (EDA) =====\n")
# Convert JSON → DataFrame (flatten useful fields)

df = pd.DataFrame([{

   "user_id": r["user_profile"]["userid"],

   "location": r["user_profile"]["location"],

   "followers": r["user_profile"]["followers"],

   "following": r["user_profile"]["following_count"],

   "caption": r["content_data"]["text_caption"],

   "language": r["content_data"]["language"],

   "likes": r["engagement_metrics"]["no_of_likes_count"],

   "comments": r["engagement_metrics"]["no_of_comments_count"],

   "shares": r["engagement_metrics"]["no_of_share_counts"]} for r in subset_data])

#     1. BASIC INFO
```

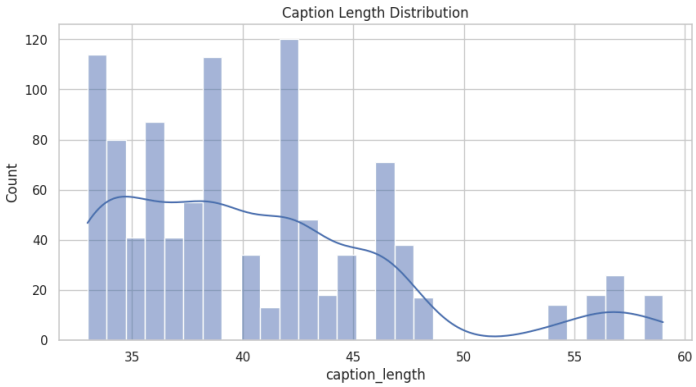
```
print("Dataset Shape:", df.shape)

print("\nColumns:\n", df.columns)

print("\nSample Data:\n", df.head())

#	2. MISSING VALUES

print("\nMissing Values:\n", df.isnull().sum())

#	3. LANGUAGE DISTRIBUTION

plt.figure(figsize=(10,5))

sns.countplot(data=df, x="language", order=df["language"].value_counts().index)

plt.title("Language Distribution")

plt.show()

#	4. ENGAGEMENT ANALYSIS

plt.figure(figsize=(10,5))

sns.histplot(df["likes"], bins=30, kde=True)

plt.title("Likes Distribution")

plt.show()
plt.figure(figsize=(10,5))

sns.histplot(df["comments"], bins=30, kde=True)

plt.title("Comments Distribution")

plt.show()
#	5. TOP LOCATIONS

plt.figure(figsize=(10,5))

sns.countplot(data=df, x="location", order=df["location"].value_counts().index)

plt.xticks(rotation=45)

plt.title("Top User Locations")

plt.show()

#	6. CORRELATION HEATMAP

plt.figure(figsize=(8,6))

sns.heatmap(df[["followers","following","likes","comments","shares"]].corr(), annot=True, cmap="coolwarm")
```

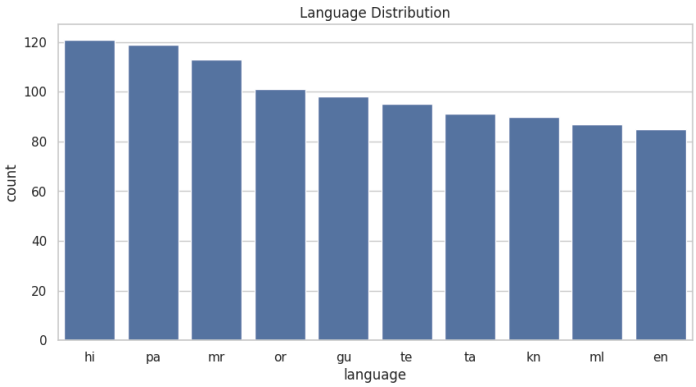
```
plt.title("Feature Correlation Heatmap")

plt.show()

#     7. CAPTION LENGTH ANALYSIS

df["caption_length"] = df["caption"].apply(len)
plt.figure(figsize=(10,5))

sns.histplot(df["caption_length"], bins=30, kde=True)

plt.title("Caption Length Distribution")

plt.show()
```

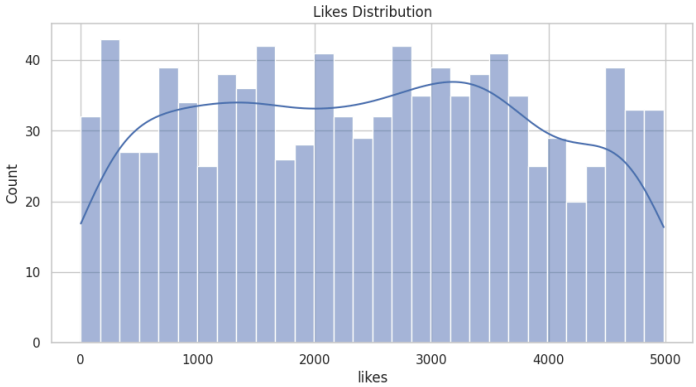
OUTPUT:

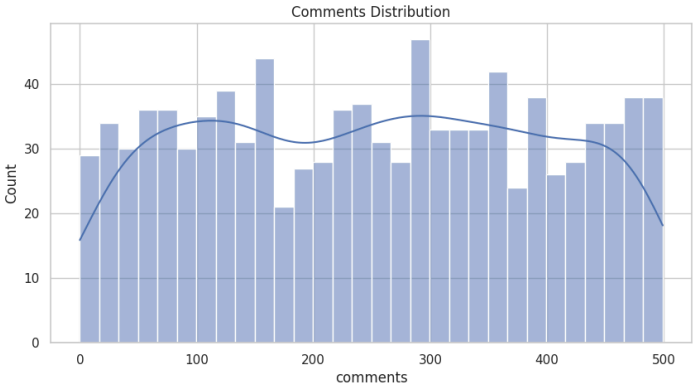
```
===== EXPLORATORY DATA ANALYSIS (EDA) =====
Dataset Shape: (1000, 9)
Columns:
 Index(['user_id', 'location', 'followers', 'following', 'caption', 'language',
       'likes', 'comments', 'shares'],
      dtype='object')
Sample Data:
      user_id   location  followers  following  \
0   user_9499     Jaipur      16184         59
1  user_38985      Delhi      68132        505
2  user_30619  Hyderabad      47614        677
3  user_95815    Chennai      34166        613
4  user_70591    Kolkata       5973        111
                                caption language  likes  comments  \
0     આજે હવામાન ખૂબ સરસ છે! #gu #socialmedia       gu   4909        78
1  आज कड़ी मेहनत कर रहा हूँ। #hi #socialmedia       hi   2417         4
2         ਸੁੰਦਰ ਸੂਰਜ ਡੁੱਬਣ! #pa #socialmedia       pa   3747        85
3        ನೈಹಿತರೊಂದಿಗೆ ಊಟ. #kn #socialmedia       kn   3517       279
4       ଆଜି ବହୁତ କାମ କରୁଛି। #or #socialmedia       or   1864       196
   shares
0     370
1     365
2     278
3     490
4       2
Missing Values:
 user_id      0
location     0
followers    0
following    0
caption      0
```

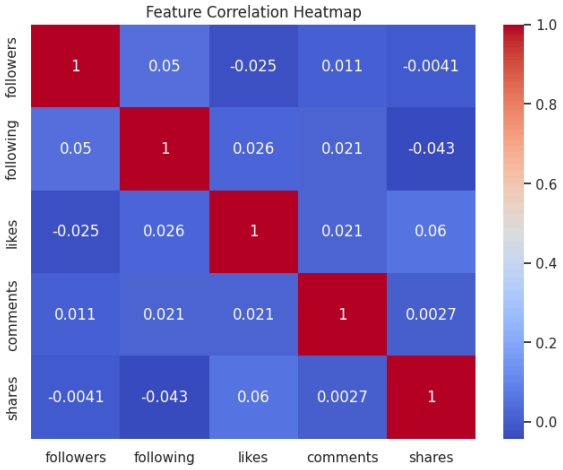
language 0
likes 0
comments 0
shares 0
dtype: int64

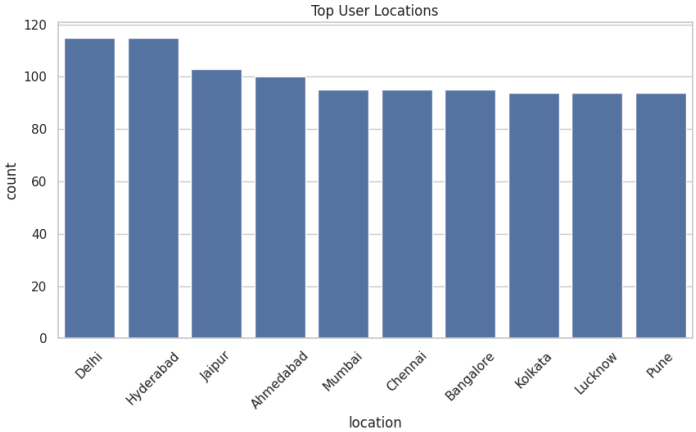












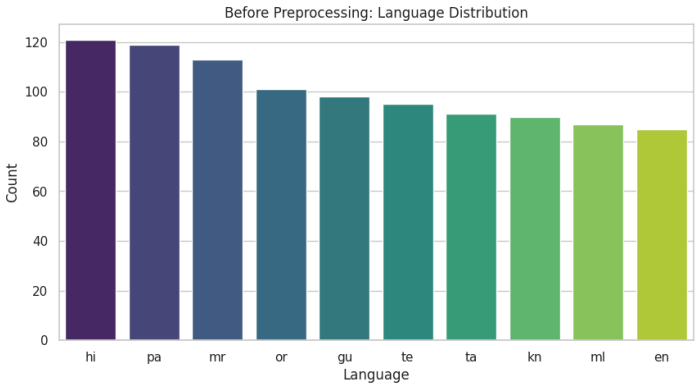
Step 4: Visualization (Before Preprocessing)

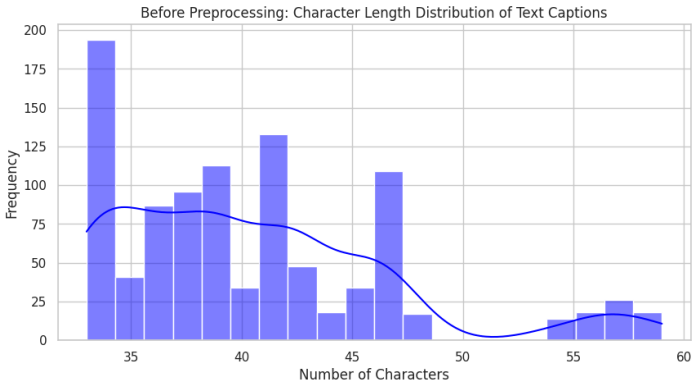
- Language count plots
- Caption length distribution

```
# Visualization 1: Language Distribution

langs = [r['content_data']['language'] for r in subset_data]

plt.figure(figsize=(10,5))

sns.countplot(x=langs, palette='viridis', order=pd.Series(langs).value_counts().index)
```

```
plt.title('Before Preprocessing: Language Distribution')

plt.xlabel('Language')

plt.ylabel('Count')

plt.show()
# Visualization 2: Character Length Distribution Before Preprocessing

original_lengths = [len(r['content_data']['text_caption']) for r in subset_data]

plt.figure(figsize=(10,5))

sns.histplot(original_lengths, bins=20, color='blue', kde=True)

plt.title('Before Preprocessing: Character Length Distribution of Text Captions')

plt.xlabel('Number of Characters')

plt.ylabel('Frequency')

plt.show()
```

OUTPUT:





Step 5: Text Preprocessing

- Removed URLs, hashtags, mentions
- Removed special characters
- Cleaned text for modeling

```
import re

import nltk

nltk.download('punkt')

nltk.download('punkt_tab')
print("Applying Text Preprocessing Pipeline...")
def clean_text(text):

    text = re.sub(r'http\S+', '', text) # Remove URLs

    text = re.sub(r'@\S+', '', text) # Remove mentions
```

```
    text = re.sub(r'#\S+', '', text) # Remove hashtags

    text = re.sub(r'[^\w\s]', '', text) # Remove punctuation

    return text.strip()
for item in subset_data:

    original = item["content_data"]["text_caption"]

    cleaned = clean_text(original)

    item["content_data"]["cleaned_caption"] = cleaned

print("Preprocessing complete. Printing first 5 preprocessed text samples:\n")

for sample in subset_data[:5]:

    print("Original:", sample["content_data"]["text_caption"])

    print("Cleaned: ", sample["content_data"]["cleaned_caption"])

    print("-" * 50)
```

Output:

```
[nltk_data] Downloading package punkt to /root/nltk_data...

[nltk_data]   Package punkt is already up-to-date!

[nltk_data] Downloading package punkt_tab to /root/nltk_data...

Applying Text Preprocessing Pipeline...

Preprocessing complete. Printing first 5 preprocessed text samples:

Original: આજે હવામાન ખૂબ સરસ છે! #gu #socialmedia

Cleaned:  આજ હવમન ખબ સરસ છ

Original: आज कड़ी मेहनत कर रहा हूँ। #hi #socialmedia

Cleaned:  आज कड महनत कर रह ह

Original: ਸੁੰਦਰ ਸੂਰਜ ਡੁੱਬਣ! #pa #socialmedia

Cleaned:  ਸਦਰ ਸਰਜ ਡਬਣ

Original: ಸ್ನೇಹಿತರೊಂದಿಗೆ ಊಟ. #kn #socialmedia

Cleaned:  ಸನಹತರದಗ ಊಟ

Original: ଆଜି ବହୁତ କାମ କରୁଛି। #or #socialmedia

Cleaned:  ଆଜ ବହତ କମ କରଛ
```

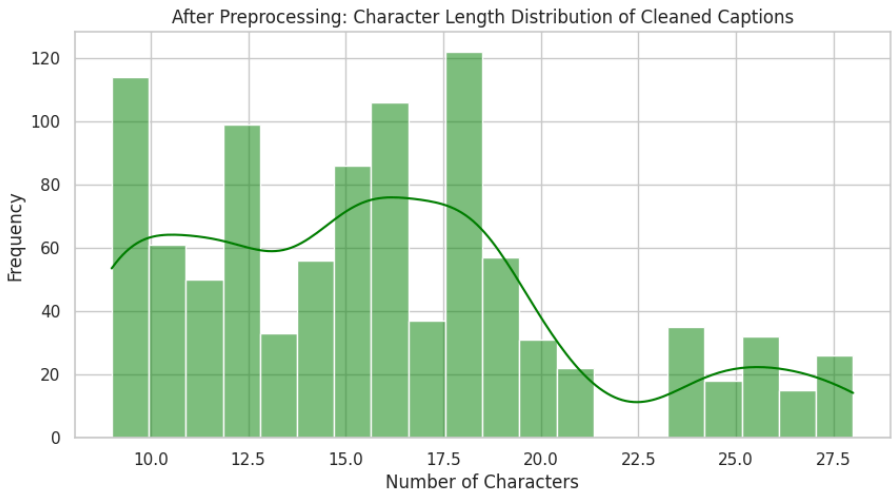
[nltk_data] Package punkt_tab is already up-to-date!

Step 6: Visualization (After Preprocessing)

- Compared text length before & after cleaning
- Observed noise reduction

```
# Visualization 3: Character Length Distribution After Preprocessing

cleaned_lengths = [len(r['content_data']['cleaned_caption']) for r in subset_data]

plt.figure(figsize=(10,5))

sns.histplot(cleaned_lengths, bins=20, color='green', kde=True)

plt.title('After Preprocessing: Character Length Distribution of Cleaned Captions')

plt.xlabel('Number of Characters')

plt.ylabel('Frequency')

plt.show()
```

Output:



Step 7: Feature Extraction

TF-IDF

- Captures importance of words

Bag of Words

- Frequency-based representation

BERT Embeddings

- Used multilingual BERT model
- Extracted semantic vectors

```
from transformers import AutoTokenizer, AutoModel
```

```
import torch

from sklearn.feature_extraction.text import TfidfVectorizer
print("Applying TF-IDF Vectorization...")
tfidf = TfidfVectorizer(max_features=5000)

tfidf_features = tfidf.fit_transform(sample_texts)
print("TF-IDF Shape:", tfidf_features.shape)
from sklearn.feature_extraction.text import CountVectorizer
print("Applying Bag of Words Vectorization...")
bow = CountVectorizer(max_features=5000)

bow_features = bow.fit_transform(sample_texts)
print("BoW Shape:", bow_features.shape)
model_name = "bert-base-multilingual-cased"

print(f"Loading Simple Multilingual BERT for Vectorization: {model_name}")
tokenizer = AutoTokenizer.from_pretrained(model_name)

model = AutoModel.from_pretrained(model_name)

if torch.cuda.is_available():

    model = model.to('cuda')
total_params = sum(p.numel() for p in model.parameters())

print(f"Total Model parameters: {total_params:,}")

print(f"Model size: {total_params * 4 / (1024**2):.2f} MB")
# Extract text and languages for vectorization

sample_texts = [r["content_data"]["cleaned_caption"] for r in subset_data[:500]]

sample_langs = [r["content_data"]["language"] for r in subset_data[:500]]
print("Applying Vectorization to extract embeddings in batches to prevent Out-Of-Memory (OOM) errors...")

batch_size = 32

all_embeddings = []
model.eval()

with torch.no_grad():

    for i in range(0, len(sample_texts), batch_size):

        batch_texts = sample_texts[i:i+batch_size]

        inputs = tokenizer(batch_texts, padding=True, truncation=True, return_tensors="pt", max_length=128)

        if torch.cuda.is_available():
```

```
        inputs = {k: v.to('cuda') for k, v in inputs.items()}
    outputs = model(**inputs)

    # Using the mean of the last hidden state as the sentence embedding

    batch_embeddings = outputs.last_hidden_state.mean(dim=1).cpu().numpy()

    all_embeddings.extend(batch_embeddings)
embeddings_array = np.array(all_embeddings)

print("Vectorization complete. Extracted Embeddings Array shape:", embeddings_array.shape)

print("Preview of first embedding (first 5 components):", embeddings_array[0][:5])

print("\n--- Feature Comparison ---")

print("TF-IDF Features:", tfidf_features.shape)

print("BoW Features:", bow_features.shape)

print("BERT Embeddings:", embeddings_array.shape)
```

Output:

```
Applying TF-IDF Vectorization...

TF-IDF Shape: (500, 160)

Applying Bag of Words Vectorization...

BoW Shape: (500, 160)

Loading Simple Multilingual BERT for Vectorization: bert-base-multilingual-cased

Loading weights: 100%

199/199 [00:01<00:00, 155.06it/s, Materializing param=pooler.dense.weight]

BertModel LOAD REPORT from: bert-base-multilingual-cased

Key                                        | Status     |  |

-------------------------------------------+------------+--+-

cls.predictions.bias                       | UNEXPECTED |  |

cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |  |

cls.predictions.transform.dense.weight     | UNEXPECTED |  |

cls.seq_relationship.weight                | UNEXPECTED |  |

cls.predictions.transform.dense.bias       | UNEXPECTED |  |
```

```
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |

cls.seq_relationship.bias                  | UNEXPECTED |  |

Notes:

- UNEXPECTED   :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Total Model parameters: 177,853,440

Model size: 678.46 MB

Applying Vectorization to extract embeddings in batches to prevent Out-Of-Memory (OOM) errors...

Vectorization complete. Extracted Embeddings Array shape: (500, 768)

Preview of first embedding (first 5 components): [ 0.5449723  -0.3152861   0.09696994  0.17954996  0.4980652 ]

--- Feature Comparison ---

TF-IDF Features: (500, 160)

BoW Features: (500, 160)

BERT Embeddings: (500, 768)
```

Step 8: Dimensionality Reduction

PCA

- Reduced feature dimensions

t-SNE

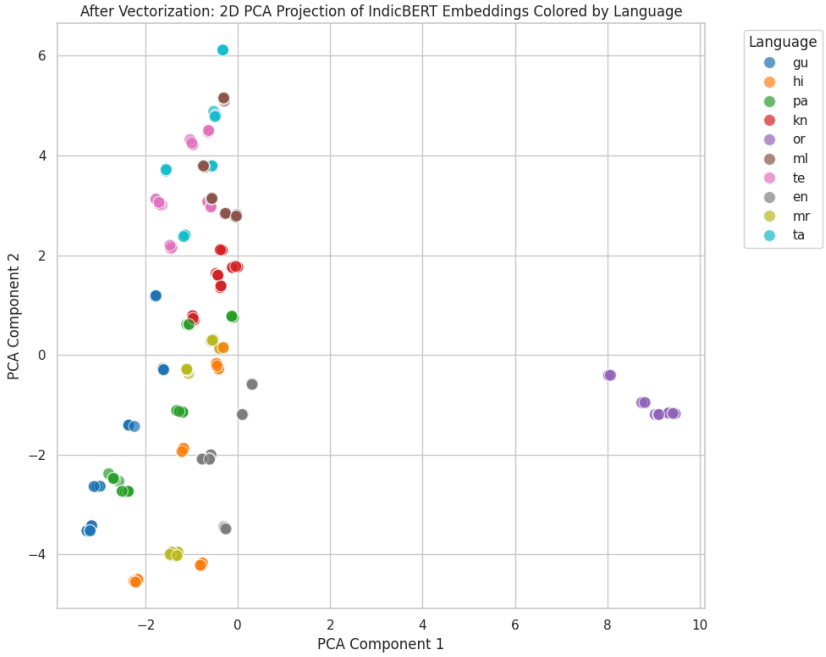
- Visualized clustering of languages

```
# Visualization 4: PCA dimensionality reduction of extracted embeddings

print("Applying PCA to reduce embeddings to 2D for visualization...")

pca = PCA(n_components=2)

embeddings_2d = pca.fit_transform(embeddings_array)
plt.figure(figsize=(10, 8))

sns.scatterplot(x=embeddings_2d[:, 0], y=embeddings_2d[:, 1], hue=sample_langs, palette='tab10', s=100, alpha=0.7)

plt.title('After Vectorization: 2D PCA Projection of IndicBERT Embeddings Colored by Language')

plt.xlabel('PCA Component 1')
```

```
plt.ylabel('PCA Component 2')

plt.legend(title='Language', bbox_to_anchor=(1.05, 1), loc='upper left')

plt.tight_layout()

plt.show()
from sklearn.manifold import TSNE
print("Applying t-SNE for dimensionality reduction...")
# Reduce to 2D

tsne = TSNE(n_components=2, perplexity=30, learning_rate=200, random_state=42)

tsne_embeddings = tsne.fit_transform(embeddings_array)
print("t-SNE transformation complete. Shape:", tsne_embeddings.shape)
# Plot

plt.figure(figsize=(10, 8))

sns.scatterplot(

    x=tsne_embeddings[:, 0],

    y=tsne_embeddings[:, 1],

    hue=sample_langs,

    palette='tab10',

    s=100,

    alpha=0.7)
plt.title("t-SNE Visualization of Text Embeddings (Colored by Language)")

plt.xlabel("t-SNE Component 1")

plt.ylabel("t-SNE Component 2")

plt.legend(title="Language", bbox_to_anchor=(1.05, 1), loc='upper left')

plt.tight_layout()

plt.show()
```

Output:

```
Applying PCA to reduce embeddings to 2D for visualization...
```



Applying t-SNE for dimensionality reduction...

t-SNE transformation complete. Shape: (500, 2)

Step 9: Classification Models and Prediction

Machine Learning Models

- Logistic Regression
- Naive Bayes
- Random Forest

Deep Learning Model

- Artificial Neural Network (ANN)
- Tested models on new text inputs
- Predicted language labels

STEP 1: Imports

```
import numpy as np

import pandas as pd
from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder

from sklearn.feature_extraction.text import TfidfVectorizer
# ML Models

from sklearn.linear_model import LogisticRegression

from sklearn.naive_bayes import MultinomialNB

from sklearn.ensemble import RandomForestClassifier
# Evaluation

from sklearn.metrics import accuracy_score, classification_report
# DL Model

import tensorflow as tf

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Dense

# STEP 2: Prepare Data
texts = [r["content_data"]["cleaned_caption"] for r in subset_data[:5000]]

labels = [r["content_data"]["language"] for r in subset_data[:5000]]
# Encode labels

le = LabelEncoder()

y = le.fit_transform(labels)
# TF-IDF Vectorization

vectorizer = TfidfVectorizer(max_features=5000)

X = vectorizer.fit_transform(texts).toarray()
# Train-Test Split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print("Data Ready:", X_train.shape)

# STEP 3: ML MODELS
models = {

    "Logistic Regression": LogisticRegression(max_iter=200),

    "Naive Bayes": MultinomialNB(),
```

```
    "Random Forest": RandomForestClassifier(n_estimators=100, random_state=42)

}
print("\n===== ML MODELS RESULTS =====\n")
for name, model in models.items():

    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)
    print(f"\n     {name}")

    print("Accuracy:", accuracy_score(y_test, y_pred))

    print(classification_report(y_test, y_pred))
# STEP 4: DEEP LEARNING MODEL
print("\n===== DEEP LEARNING MODEL =====\n")
dl_model = Sequential([

    Dense(256, activation='relu', input_shape=(X_train.shape[1],)),

    Dense(128, activation='relu'),

    Dense(len(le.classes_), activation='softmax')

])
dl_model.compile(

    optimizer='adam',

    loss='sparse_categorical_crossentropy',

    metrics=['accuracy']

)
history = dl_model.fit(

    X_train, y_train,

    epochs=5,

    batch_size=32,

    validation_split=0.1,

    verbose=1

)
# Evaluate DL Model

loss, acc = dl_model.evaluate(X_test, y_test)

print("\nDL Model Accuracy:", acc)
```

```
# STEP 5: TEST WITH NEW INPUT
print("\n===== TEST SAMPLE =====\n")
new_text = ["आज मौसम बहुत अच्छा है"]
new_vector = vectorizer.transform(new_text).toarray()
# Test with Random Forest (best usually)

rf_model = models["Random Forest"]

prediction = rf_model.predict(new_vector)
print("Input Text:", new_text[0])

print("Predicted Language:", le.inverse_transform(prediction)[0])
```

Output:

```
Data Ready: (800, 160)
===== ML MODELS RESULTS =====
    Logistic Regression
Accuracy: 0.965
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        14
           1       1.00      1.00      1.00        22
           2       0.82      1.00      0.90        31
           3       1.00      1.00      1.00        19
           4       1.00      1.00      1.00        13
           5       1.00      0.71      0.83        24
           6       1.00      1.00      1.00        14
           7       1.00      1.00      1.00        29
           8       1.00      1.00      1.00        15
           9       1.00      1.00      1.00        19
    accuracy                           0.96       200
   macro avg       0.98      0.97      0.97       200
weighted avg       0.97      0.96      0.96       200
    Naive Bayes
Accuracy: 0.965
              precision    recall  f1-score   support
           0       1.00      1.00      1.00        14
           1       1.00      1.00      1.00        22
           2       0.82      1.00      0.90        31
           3       1.00      1.00      1.00        19
           4       1.00      1.00      1.00        13
           5       1.00      0.71      0.83        24
           6       1.00      1.00      1.00        14
           7       1.00      1.00      1.00        29
           8       1.00      1.00      1.00        15
           9       1.00      1.00      1.00        19
```

```
   accuracy                         0.96      200
  macro avg       0.98      0.97      0.97      200
weighted avg       0.97      0.96      0.96      200
   Random Forest
Accuracy: 0.965
              precision    recall  f1-score   support
           0       1.00      1.00      1.00        14
           1       1.00      1.00      1.00        22
           2       0.82      1.00      0.90        31
           3       1.00      1.00      1.00        19
           4       1.00      1.00      1.00        13
           5       1.00      0.71      0.83        24
           6       1.00      1.00      1.00        14
           7       1.00      1.00      1.00        29
           8       1.00      1.00      1.00        15
           9       1.00      1.00      1.00        19
   accuracy                         0.96      200
  macro avg       0.98      0.97      0.97      200
weighted avg       0.97      0.96      0.96      200
===== DEEP LEARNING MODEL =====

Epoch 1/5

23/23 ━━━━━━━━━━━━━━━━━━━━ 6s 141ms/step - accuracy: 0.6958 - loss: 2.1172 - val_accuracy: 0.9625 - val_loss: 1.8781

Epoch 2/5

23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9833 - loss: 1.4959 - val_accuracy: 0.9625 - val_loss: 1.0438

Epoch 3/5

23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9833 - loss: 0.5544 - val_accuracy: 0.9625 - val_loss: 0.2390

Epoch 4/5

23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9764 - loss: 0.1105 - val_accuracy: 0.9625 - val_loss: 0.0851

Epoch 5/5

23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9806 - loss: 0.0497 - val_accuracy: 0.9750 - val_loss: 0.0595

7/7 ━━━━━━━━━━━━━━━━━━━━ 1s 113ms/step - accuracy: 0.9750 - loss: 0.0565

DL Model Accuracy: 0.9750000238418579

===== TEST SAMPLE =====
Input Text: आज मौसम बहुत अच्छा है
Predicted Language: hi
```

Step 11: Network Graph Visualization

- Built graph using NetworkX
- Visualized user connections
- Identified influential users using centrality

```
# Visualization 6: Advanced Network Graph Visualization
print("Visualizing Network Graph (User-Friend Connections)...")
G = nx.Graph()

# Add edges for top 50 users to keep graph readable
for record in subset_data[:50]:
    user = record['user_profile']['userid']
    connections = record['network_relationship_data']['follow_friend_connection']
    for friend in connections:
        G.add_edge(user, friend)

plt.figure(figsize=(12, 10))
pos = nx.spring_layout(G, k=0.15)
nx.draw_networkx_nodes(G, pos, node_size=30, node_color='teal', alpha=0.7)
nx.draw_networkx_edges(G, pos, alpha=0.2, edge_color='gray')
plt.title('Network Relationship Graph (Users & Friends)')
plt.axis('off')
plt.show()

print("Building Advanced Social Network Graph...")

G = nx.Graph()

# Build graph
for record in subset_data[:100]:
    user = record['user_profile']['userid']
    connections = record['network_relationship_data']['follow_friend_connection']

    for friend in connections:
        G.add_edge(user, friend)

# Calculate centrality
degree_centrality = nx.degree_centrality(G)

# Get top influencers
top_users = sorted(degree_centrality.items(), key=lambda x: x[1], reverse=True)[:5]

print("\nTop Influential Users:")
for user, score in top_users:
    print(f"{user} → Centrality Score: {score:.4f}")
plt.figure(figsize=(12, 10))

pos = nx.spring_layout(G, k=0.2)

# Node size based on importance
node_sizes = [v * 2000 for v in degree_centrality.values()]
```

```
nx.draw_networkx_nodes(G, pos, node_size=node_sizes, node_color='skyblue', alpha=0.7)
nx.draw_networkx_edges(G, pos, alpha=0.3)

plt.title("Enhanced Social Network Graph (Node size = Influence)")
plt.axis('off')
plt.show()
```

OUTPUT:

Visualizing Network Graph (User-Friend Connections)...

Building Advanced Social Network Graph...

Top Influential Users:

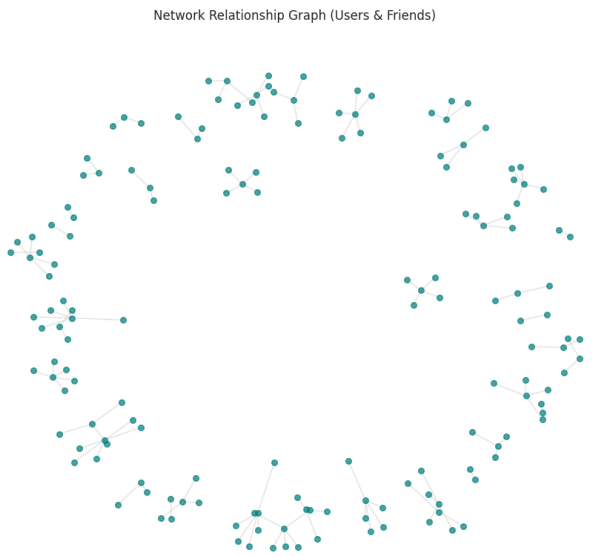
user_15874 → Centrality Score: 0.0145

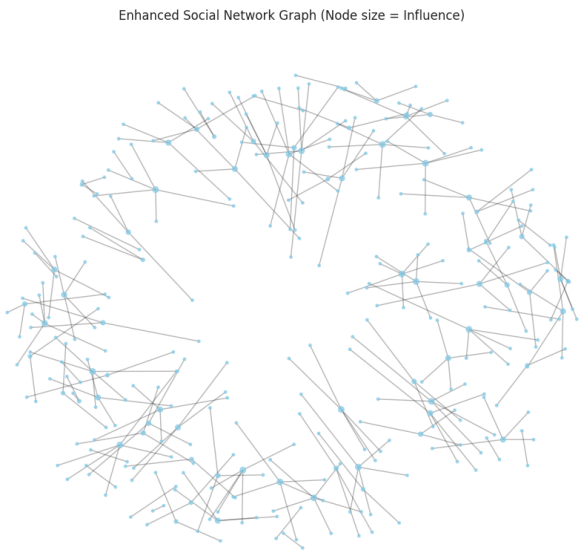
user_84905 → Centrality Score: 0.0145

user_41989 → Centrality Score: 0.0145

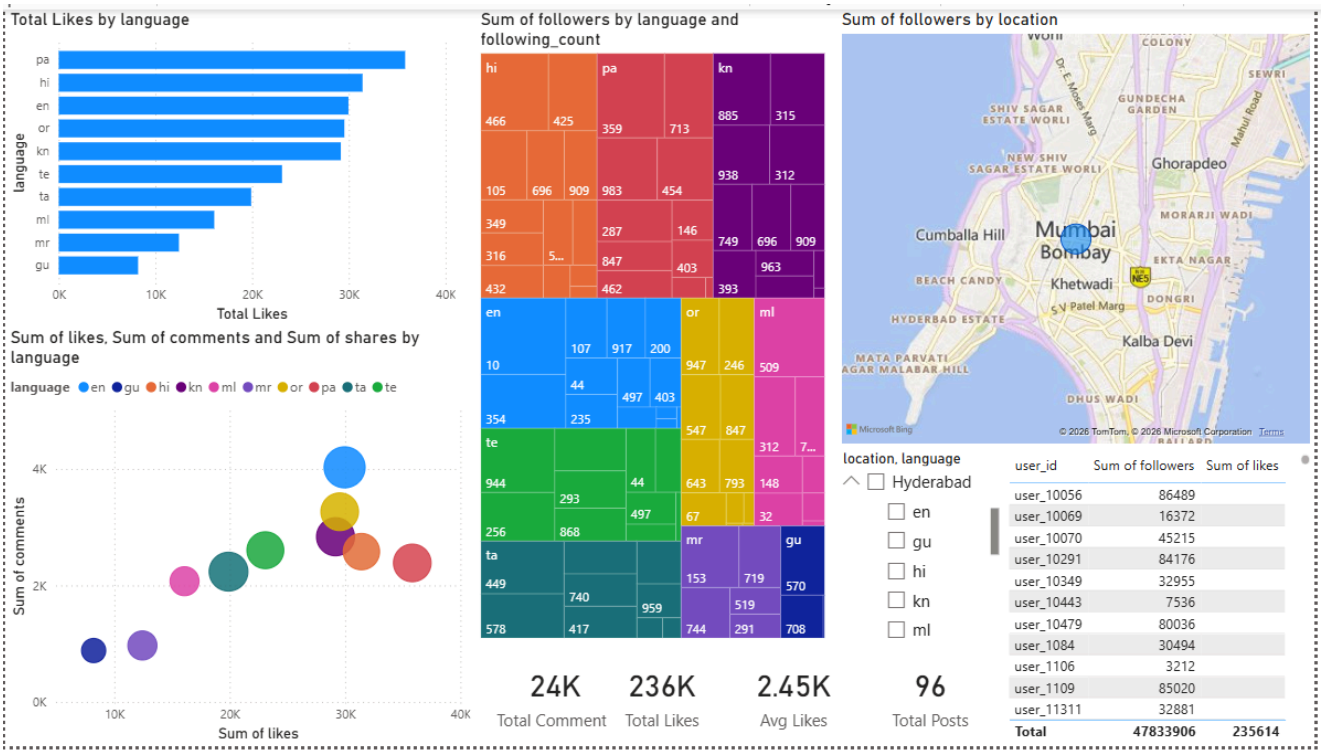
user_13993 → Centrality Score: 0.0145

user_71451 → Centrality Score: 0.0145





Dashboard:



Result

The data analysis pipeline was successfully implemented on a synthetic multilingual social network dataset with 500,000 records. The system performed data generation, preprocessing, feature extraction (TF-IDF, BoW, BERT), dimensionality reduction (PCA, t-SNE), classification, and network graph analysis.

All models performed well, with Machine Learning models achieving around 96.5% accuracy and the Deep Learning model achieving around 97.5% accuracy. The system also successfully predicted languages for new inputs and visualized user relationships using graph analysis.

Conclusion

The project demonstrates an end-to-end NLP and machine learning pipeline for analyzing multilingual social media data. It shows that combining traditional ML, deep learning, and graph analytics can effectively extract meaningful insights from large-scale user-generated content and network interactions.